Non Functional Requirement

Prompt

Act as a Software Architect and define the non-functional requirements (NFRs) for the Inventory Management System (IMS), considering these constraints:

- **Product catalog:** 10K–500K products
- **Geographic coverage:** India, Japan, and the US, with potential EU expansion
- **Users:** 5K–15K
- **Throughput:** Up to 500 TPS per region

Cover performance, scalability, availability, reliability, security, data consistency, observability, disaster recovery, and regional deployment. Define measurable targets where possible, state assumptions, and identify requirements needing clarification.

Response

Non-Functional Requirements (NFRs) – Inventory Management System

- **Scalability** – The system shall support product catalogs ranging from 10K to 500K products and scale horizontally to support future regional expansion including India, Japan, the US, and EU markets.
- **Performance** – The system shall support up to 500 transactions per second (TPS) per region with standard API response times below 2–3 seconds for normal operations and under 5 seconds for heavy queries.
- **Concurrent User Support** – The platform shall support 5K–15K active users across regions without degradation in performance or availability.
- **High Availability** – The system shall maintain at least 99.9% uptime with regional failover and redundancy mechanisms to minimize downtime.
- **Database Performance** – The database architecture shall support high-volume inventory reads/writes, optimized indexing, partitioning, and efficient query execution for large catalogs and transaction histories.

- **Multi-Region Support** – The system shall support deployment across multiple geographic regions with localized configurations, regional data segregation, and low-latency access.
- **Localization & Internationalization** – The platform shall support multiple languages, currencies, date/time formats, measurement units, and timezone handling for India, Japan, the US, and future EU regions.
- **Data Consistency** – Inventory transactions shall maintain strong consistency and ACID-compliant transactional integrity to prevent stock mismatches during concurrent operations.
- **Concurrency Handling** – The system shall safely manage simultaneous stock updates using locking, optimistic concurrency, or distributed transaction strategies.
- **Security** – All sensitive data shall be encrypted in transit and at rest, with RBAC, MFA support, secure APIs, audit trails, and compliance with regional security standards.
- **Compliance** – The platform shall support compliance requirements such as GDPR (EU), regional data retention policies, audit logging, and access traceability.
- **Disaster Recovery & Backup** – Automated backups, point-in-time recovery, and disaster recovery procedures shall ensure recovery objectives such as RPO < 15 minutes and RTO < 1 hour.
- **Monitoring & Observability** – The system shall include centralized logging, distributed tracing, real-time monitoring, alerting, and operational dashboards for infrastructure and application health.
- **Reliability** – All inventory operations shall support retry mechanisms, rollback handling, idempotency protection, and fault tolerance to avoid duplicate or inconsistent transactions.
- **Maintainability** – The application shall use modular architecture, coding standards, API versioning, and automated CI/CD pipelines to simplify maintenance and future enhancements.
- **Extensibility** – The architecture shall support future integrations including ERP systems, barcode/RFID solutions, mobile applications, AI forecasting, and additional warehouse regions.
- **Usability** – The UI shall be responsive, intuitive, and optimized for warehouse and operations users with minimal training requirements.
- **Search Performance** – Product searches and inventory lookups across catalogs up to 500K products shall return results within acceptable response times (typically <2 seconds).
- **Bulk Operation Handling** – The system shall efficiently process large imports, exports, and bulk inventory updates with asynchronous/background job processing.
- **Auditability** – All inventory-affecting actions shall be fully traceable with immutable audit logs capturing user, timestamp, action type, and before/after values.
- **API Reliability** – APIs shall support throttling, rate limiting, idempotency, and backward compatibility for external integrations and high-volume consumers.
- **Deployment & Infrastructure** – The platform shall support cloud-native deployment, containerization, auto-scaling, and infrastructure-as-code for multi-region operations.

- **Data Retention & Archival** – Historical inventory and transaction data shall support configurable archival strategies without impacting operational performance.
- **Network Resilience** – The system shall gracefully handle temporary network failures, retries, and offline synchronization scenarios in warehouse operations.
- **Accessibility** – The UI should follow accessibility standards such as WCAG guidelines to support usability across different user groups.